CT · WEEK 3

CT week 3 patterns: Procedures, slabs and multiple passes

Procedures, parameters & return · Slab and bill problems (if / else if) · Multiple (non-nested) iterations. Save this PDF for revision.

PROCEDURES, PARAMETERS & RETURN · IN EVERY PAPER

A procedure

```
Procedure isVowel(c)
    if(c == 'a' or c == 'e' or c == 'i' or c == 'o' or c == 'u'){
        return(True)
    }
    return(False)
End isVowel
```

PROCEDURES, PARAMETERS & RETURN · IN EVERY PAPER

Key points

- Parameters are the inputs. A call like isVowel(letter) substitutes the argument for c.
- return(v) hands v back and exits immediately. Nothing after it runs.
- Side effect: the procedure changes something outside itself (moves rows between tables, changes an outside variable). Ask whether the table is the same after the call.
- Loop counting: i = 1; while(i <= 20){ i = i * 2 } runs for i = 1, 2, 4, 8, 16: five times.

PROCEDURES, PARAMETERS & RETURN · IN EVERY PAPER

Standard procedures: learn the trick

Task	Trick
Sum of the two smallest of a, b, c (PYQ Jan 2024)	a + b + c − max(a, b, c)
Largest of three	compare a with b, then the winner with c
Even or odd	n mod 2 == 0 ⇒ even
Last digit / drop it	n mod 10 / floor(n / 10)

PROCEDURES, PARAMETERS & RETURN · IN EVERY PAPER

Standard procedures: learn the trick

Task	Trick
Sum of digits	repeat s = s + n mod 10, n = floor(n / 10) until n = 0
Reverse a number	repeat r = r × 10 + n mod 10, n = floor(n / 10)
GCD (Euclid)	while(b != 0){ t = b; b = a mod b; a = t } → answer a
Divisible by k	n mod k == 0

SLAB AND BILL PROBLEMS (IF / ELSE IF) · OFTEN ASKED

₹5 for the first 100 units, ₹7 for the next 100, ₹10 after that

```
if(unit <= 100){
    cost = unit * 5
}
else if(unit <= 200){
    cost = 100 * 5 + (unit - 100) * 7
}
else{
    cost = 100 * 5 + 100 * 7 + (unit - 200) * 10
}
```

SLAB AND BILL PROBLEMS (IF / ELSE IF) · OFTEN ASKED

Key points

- Each rate applies only to the units inside its own slab.
- An else-if chain checks top to bottom and stops at the first true condition, so the order of the checks matters.

SLAB AND BILL PROBLEMS (IF / ELSE IF) · OFTEN ASKED

Common traps

⚠ Wrong-option pattern: cost = unit * 10 for unit > 200, which charges every unit at the top rate.

⚠ Boundaries: is exactly 100 units in slab 1 (<=) or slab 2 (<)? Check the comparison sign.

MULTIPLE (NON-NESTED) ITERATIONS · OFTEN ASKED

PYQ (Jan 2024): students above the class average

```
sum = 0
n = 0
while(Table 1 has more rows){
    Read the first row X in Table 1
    sum = sum + X.Total
    n = n + 1
    Move X to Table 2
}
avg = sum / n
Move all rows from Table 2 to Table 1
count = 0
while(Table 1 has more rows){
    Read the first row X in Table 1
    if(X.Total > avg){
        count = count + 1
    }
    Move X to Table 2
}
```

MULTIPLE (NON-NESTED) ITERATIONS · OFTEN ASKED

Key points

- You can't compare with the average until you know it, so it takes two passes.
- Restore Table 1 between passes, or the second loop runs zero times.
- Three prizes problem: keep first, second and third in one pass, shifting values down whenever a bigger total arrives.

Notes for every week of CT, free.

Full notes, the pattern sheet and common traps, with worked PYQs.

btechi.in/qualifier/notes/ct →

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